

Overfitting in Wrapper-Based Feature Subset Selection: The Harder You Try the Worse it Gets

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Abstract. In Wrapper based feature selection, the more states that are visited during the search phase of the algorithm the greater the likelihood of finding a feature subset that has a high internal accuracy while generalizing poorly. When this occurs, we say that the algorithm has overfitted to the training data. We outline a set of experiments to show this and we introduce a modified genetic algorithm to address this overfitting problem by stopping the search before overfitting occurs. This new algorithm called GAWES (Genetic Algorithm With Early Stopping) reduces the level of overfitting and yields feature subsets that have a better generalization accuracy.

1 Introduction

The benefits of wrapper-based techniques for feature selection are well established [1, 15]. However, it has recently been recognized that wrapper-based techniques have the potential to overfit the training data [2]. That is, feature subsets that perform well on the training data may not perform as well on data not used in the training process. Furthermore, the extent of the overfitting is related to the depth of the search. Reunanen [2] shows that, whereas Sequential Forward Floating Selection (SFFS) beats Sequential Forward Selection on the data used in the training process, the reverse is true on hold-out data. He argues that this is because SFFS is a more intensive search process i.e. it explores more states.

In this paper we present further evidence of this and explore the use of the number of states explored in the search as an indicator of the *depth* of the search and thus as a predictor of overfitting. Clearly this metric does not tell the whole story since for example a lengthy random search will not overfit at all.

We also explore a solution to this overfitting problem. Techniques from Machine Learning research for tackling overfitting include:

- **Post-Pruning:** Overfitting can be eliminated by pruning as is done in the construction of Decision Trees [6].

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- **Jitter:** Adding noise to the training data can make it more difficult for the learning algorithm to fit the training data and thus overfitting is avoided [12].
- **Early Stopping:** Overfitting is avoided in the training of supervised Neural Networks by stopping the training when performance on a validation set starts to deteriorate [7, 14].

Of these three options, the one that we explore here is Early Stopping. We present a stochastic search process that has a cross-validation stage to determine when overfitting occurs. Then the final search uses all the data to guide the search and stops at this point determined by the cross-validation. We show that this method works well in reducing the overfitting associated with feature selection – this will be shown later in Section 4. In Section 2 of the paper we briefly discuss different approaches to Feature Selection, focusing on various wrapper based search strategies. Section 3 provides more detail on the GAWES algorithm and early stopping in stochastic search. Section 4 outlines the results of the experimental study. Future avenues for research are discussed in Section 5 and the paper concludes in Section 6.

2 Wrapper-Based Feature Subset Selection

Feature selection is defined as the selection of a subset of features to describe a phenomenon from a larger set that may contain irrelevant or redundant features. Improving classifier performance and accuracy are usually the motivating factors behind this, as the accuracy is degraded by the presence of these irrelevant features. The *curse of dimensionality* is the term given to the phenomenon when there are too many features in the model and not enough instances to completely describe the target concept. Feature selection attempts to identify and eliminate unnecessary features, thereby reducing the dimensionality of the data, and hopefully resulting in an increase in accuracy.

The two common approaches to feature selection are the use of filters and the wrapper method. Filtering techniques attempt to identify features that are related to or predictive of the outcome of interest: they operate independently of the learning algorithm. An example is Information Gain, which was originally introduced to Machine Learning research by Quinlan as a criterion for building concise decision trees [6] but it is now widely used for feature selection in general. The wrapper approach differs in that it evaluates subsets based upon the accuracy estimates provided by a classifier built with that feature subset. Thus wrappers are much more computationally expensive than filters but can produce better results because they take the *bias* of the classifier into account and evaluate features in context. A detailed presentation of the wrapper approach can be found in [1].

2.1 Search Algorithms

The wrapper method can be viewed as a search optimization process and therefore can incur a high computational cost. From n features, the number of possible feature subsets is 2^n , so it is impractical to search the whole state space except in situations